Shaping machine was designed by James Nasmith, an Englishman in the year 1936. He designed it to produce flat surface. After it was designed by him to produce various surfaces like horizontal, vertical, inclined, concave and convex surfaces.



James Nasmith

## 3.1 Introduction

Shaping is a process of machining a flat surface which may be horizontal, vertical, inclined, concave or convex using a reciprocating single point tool. A shaping machine is a reciprocating type of machine tool. James Nasmith, an Englishman designed a shaping machine to produce flat surfaces in the year 1836.

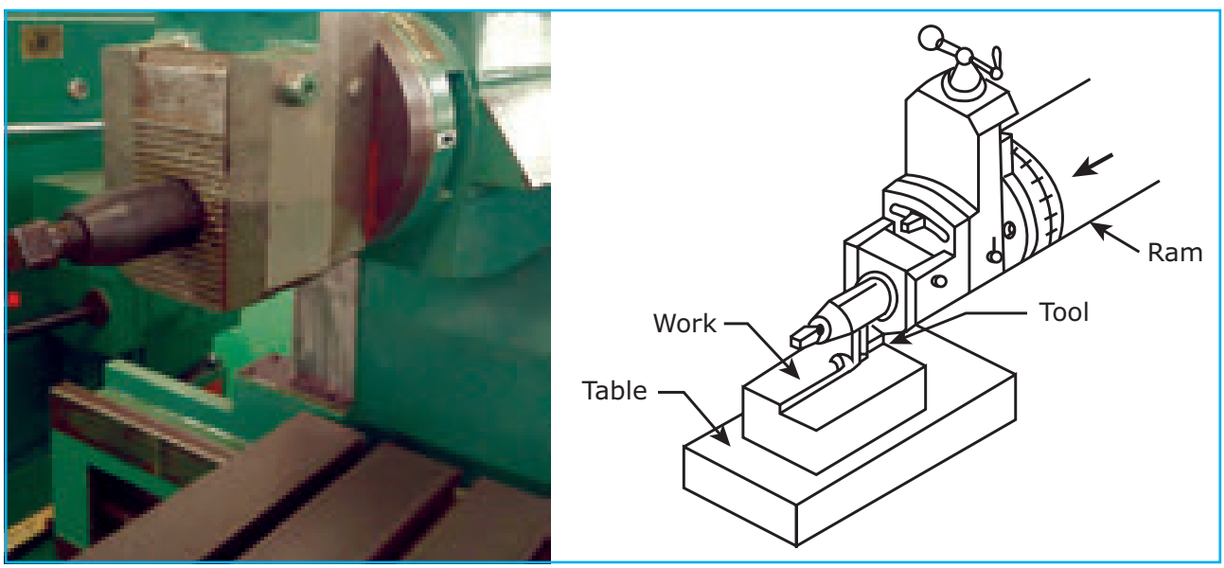
## 3.2 Process of Machining

The work is held firmly on the table and the ram is allowed to reciprocate over it. A single point cutting tool is attached to the ram. When the ram moves horizontally in the forward direction, the tool removes metal from the work. On the return stroke, metal is not removed. The ram moves at a slow speed during forward stroke. But during the forward and return stroke remain the same, the time taken by the return stroke is less as it is faster. It is possible by 'Quick return mechanism'.

In a shaping machine, a flat horizontal surface is machined by moving the work mounted on the table in a cross direction to the tool movement. When vertical surfaces are machined, the feed is given to the tool.

When a inclined surface is machined, the vertical slide of the toolhead is swiveled to the required angle and the feed is given to the tool by rotating the downfeed hand wheel.

The method of machining in a shaper is illustrated in Figure.




Machining process



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